SIMULATION EXPERIMENT – 9

❖ Computer-Aided Design and Optimization of a Microwave Impedance-Matching Network and Frequency-Selective Filter

➔ CO Mapping: CO3

Bloom's Level: K6 – Create

WK Level: WK4 – Matching/filter design

➔ Objectives of this Experiment:

1. To design an impedance-matching network for a specified complex microwave load.
2. To optimize the matching network for minimum reflection and maximum power transfer.
3. To design a microwave filter according to specified cutoff frequency, bandwidth and impedance requirements.
4. To evaluate return loss, insertion loss, passband, stopband and bandwidth using simulated frequency responses.

➔ Expected Outcome from this Experiment:

1. Design and develop an impedance-matching network for a specified complex microwave load.
2. Optimize and validate the matching network for minimum reflection and maximum

power transfer based on the simulated S-parameter response.
3. Design and simulate a microwave filter satisfying the specified cutoff frequency, bandwidth and impedance requirements.
4. Evaluate and interpret return loss, insertion loss, passband, stopband and bandwidth from the simulated frequency response and compare the obtained values with the specified design requirements.

➔ Software Required: MATLAB / ADS / HFSS / CST / Smith Chart Tool

➔ C. INPUT PARAMETER TABLE – VALUES TO BE

➔ Experimental Procedure:

A. Impedance Matching: A load ZL = 75 + j25 Ω must be matched to Z0 = 50 Ω at 10 GHz. Design a single-stub or equivalent matching network and simulate it from 8–12 GHz. Calculate the theoretical |ΓL| = |(ZL−Z0)/(ZL+Z0)|, return loss and matched input impedance. Compare theoretical and simulated values before and after matching.

B. Filter Design: Design a 50 Ω frequency-selective filter with cutoff frequency 10 GHz and evaluate it from 8–12 GHz. Obtain S11 and S21, determine the −3 dB cutoff frequency, passband, stopband and insertion loss. Compare the measured/simulated cutoff frequency with 10 GHz and calculate percentage deviation.

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%% SIMULATION EXPERIMENT 9 - Impedance Matching + Frequency-Selective Filter
%  ECA13 Microwave Engineering | CO3 | K6
%  A: Single shunt open-stub matching of ZL = 75+j25 ohm to 50 ohm @ 10 GHz
%  B: 5th-order Chebyshev (0.1 dB) low-pass filter, -3 dB cutoff = 10 GHz
clear; clc; close all;

%% ---------- INPUT PARAMETERS ----------
f0   = 10e9;          % design frequency
Z0   = 50;            % reference impedance
ZL   = 75 + 25j;      % load impedance
fc   = 10e9;          % filter cutoff (-3 dB)
f    = linspace(8e9, 12e9, 2001);   % sweep 8-12 GHz

RLmin  = 15;          % dB  (return-loss requirement)
ILmax  = 2;           % dB  (insertion-loss requirement)
Astop  = 10;          % dB  (stopband attenuation criterion)
Lar    = 0.1;         % dB  (Chebyshev passband ripple)
N      = 5;           % filter order

%% ---------- PART A : IMPEDANCE MATCHING ----------
% Theory (before matching)
GL     = (ZL - Z0)/(ZL + Z0);
absGL  = abs(GL);
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RL_bef  = -20*log10(absGL);
VSWR_b  = (1 + absGL)/(1 - absGL);

% Analytical single-stub design (shunt open-circuit stub, same Z0)
yL = Z0/ZL;  g = real(yL);  b = imag(yL);
t  = (b + [1 -1]*sqrt(g*((1-g)^2 + b^2)))/(g - 1);  % tan(beta*d), 2 solutions
d_all = atan(t)/(2*pi);   d_all(d_all<0) = d_all(d_all<0) + 0.5;   % in wavelengths
l_all = zeros(1,2);
for k = 1:2
    yin   = (yL + 1j*t(k))/(1 + 1j*yL*t(k));    % admittance at stub plane
    bs    = imag(yin);                          % must be cancelled by stub
    l_all(k) = atan(-bs)/(2*pi);                % open stub: y = j tan(beta*l)
    if l_all(k) < 0, l_all(k) = l_all(k) + 0.5; end
end
[~, ks] = min(d_all + l_all);                   % choose shortest solution
d_th = d_all(ks);  l_th = l_all(ks);

% Practical (rounded) first guess -> then optimise
p_init = round([d_th l_th], 2);                 % rounded to 0.01 lambda
cost   = @(p) abs(netGamma(f0, f0, ZL, Z0, p(1), p(2)));
p_opt  = fminsearch(cost, p_init, optimset('TolX',1e-10,'TolFun',1e-12));

% Responses
[Zin_un , G_un ] = netGamma(f, f0, ZL, Z0, 0,        0);        % unmatched
[Zin_in , G_in ] = netGamma(f, f0, ZL, Z0, p_init(1), p_init(2)); % initial
[Zin_op , G_op ] = netGamma(f, f0, ZL, Z0, p_opt(1),  p_opt(2));  % optimised
RL_un = -20*log10(abs(G_un));
RL_in = -20*log10(abs(G_in));
RL_op = -20*log10(abs(G_op));
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[~, i0] = min(abs(f - f0));

Zin_a  = Zin_op(i0);   G_a = abs(G_op(i0));
RL_aft = -20*log10(max(G_a, 1e-12));
below15 = f(RL_op >= RLmin);
BW_match = max(below15) - min(below15);          % bandwidth with RL >= 15 dB

%% ---------------- PART B : FILTER DESIGN ----------------
% Chebyshev prototype (odd order, g_{N+1} = 1)
beta_c = log(1/tanh(Lar/17.37));
gam    = sinh(beta_c/(2*N));
ak = zeros(1,N); bk = zeros(1,N); gk = zeros(1,N);
for k = 1:N
   ak(k) = sin((2*k-1)*pi/(2*N));
   bk(k) = gam^2 + sin(k*pi/N)^2;
end
gk(1) = 2*ak(1)/gam;
for k = 2:N
   gk(k) = 4*ak(k-1)*ak(k)/(bk(k-1)*gk(k-1));
end
% Place the -3 dB point at fc (ripple-edge frequency fp is lower)
eps_r = sqrt(10^(Lar/10) - 1);
fp    = fc / cosh(acosh(1/eps_r)/N);
wp    = 2*pi*fp;
Cval  = gk(1:2:end)/(Z0*wp);        % shunt capacitors (g1,g3,g5)
Lval  = gk(2:2:end)*Z0/wp;          % series inductors (g2,g4)

% ABCD cascade: C1 - L2 - C3 - L4 - C5
S11 = zeros(size(f)); S21 = zeros(size(f));
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for n = 1:numel(f)
    w = 2*pi*f(n);  M = eye(2);
    for k = 1:N
        if mod(k,2)==1
            M = M*[1 0; 1j*w*Cval((k+1)/2) 1];
        else
            M = M*[1 1j*w*Lval(k/2); 0 1];
        end
    end
    A=M(1,1); B=M(1,2); C=M(2,1); D=M(2,2);
    den = A + B/Z0 + C*Z0 + D;
    S21(n) = 2/den;
    S11(n) = (A + B/Z0 - C*Z0 - D)/den;
end
S21dB = 20*log10(abs(S21));  S11dB = 20*log10(abs(S11));
IL = -S21dB;   RLf = -S11dB;

% -3 dB cutoff (linear interpolation at first crossing)
th  = -10*log10(2);
ic  = find(S21dB <= th, 1);
fc_sim = interp1(S21dB(ic-1:ic), f(ic-1:ic), th);
dev_fc = abs(fc_sim - fc)/fc*100;

% Passband (IL <= 2 dB and RL >= 15 dB), stopband (|S21| <= -Astop dB)
pm = (IL <= ILmax) & (RLf >= RLmin);
fL_pb = f(find(pm,1));  fH_pb = f(find(pm,1,'last'));
BW_pb = fH_pb - fL_pb;
BW_3dB_in_sweep = fc_sim - f(1);                % -3 dB edge minus sweep start
fs_start = f(find(S21dB <= -Astop,1));
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IL_pb_max = max(IL(pm));   RL_pb_min = min(RLf(pm));
IL_at_8 = IL(1);

%% ---------- ONE FIGURE WITH ALL PLOTS ----------
figure('Name','Experiment 9','Color','w','Position',[50 50 1400 800]);
GHz = f/1e9;

% (1) Return loss before / after matching
subplot(2,3,1);
plot(GHz, RL_un,'r','LineWidth',1.5); hold on;
plot(GHz, min(RL_in,60),'g--','LineWidth',1.5);
plot(GHz, min(RL_op,60),'b','LineWidth',1.8);
yline(RLmin,'k:','15 dB'); xline(10,'k:');
grid on; xlabel('Frequency (GHz)'); ylabel('Return loss (dB)');
title('Matching: Return Loss'); ylim([0 60]);
legend('Before matching','Initial stub','Optimised stub','Location','north');

% (2) Smith chart
subplot(2,3,2); hold on; axis equal off; smithGrid();
gz = G_un; plot(real(gz),imag(gz),'r','LineWidth',1.5);
gz = G_op; plot(real(gz),imag(gz),'b','LineWidth',1.8);
plot(real(G_un(i0)),imag(G_un(i0)),'ro','MarkerFaceColor','r');
plot(real(G_op(i0)),imag(G_op(i0)),'bs','MarkerFaceColor','b');
title('Smith Chart (8-12 GHz)');
legend("","","","","","","","Before','After','@10 GHz (before)','@10 GHz (after)', ...
      'Location','southoutside','Orientation','horizontal');

% (3) Input impedance
subplot(2,3,3);
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plot(GHz, real(Zin_un),'r-', GHz, imag(Zin_un),'r--', ...
    GHz, real(Zin_op),'b-', GHz, imag(Zin_op),'b--','LineWidth',1.4);
grid on; xline(10,'k:'); xlabel('Frequency (GHz)'); ylabel('Z_{in} (\Omega)');
title('Input Impedance'); ylim([-60 140]);
legend('Re (before)','Im (before)','Re (after)','Im (after)','Location','best');

% (4) Filter S11 and S21
subplot(2,3,4);
plot(GHz, S21dB,'b','LineWidth',1.8); hold on;
plot(GHz, S11dB,'r','LineWidth',1.5);
yline(th,'k:','-3 dB'); xline(fc/1e9,'k:');
plot(fc_sim/1e9, th,'ko','MarkerFaceColor','y');
grid on; xlabel('Frequency (GHz)'); ylabel('Magnitude (dB)');
title('Filter S_{21} and S_{11}'); ylim([-45 2]);
legend('S_{21}','S_{11}','Location','southwest');

% (5) Passband zoom: IL and RL with limits
subplot(2,3,5);
yyaxis left;  plot(GHz, IL,'b','LineWidth',1.6); ylabel('Insertion loss (dB)');
hold on; yline(ILmax,'b:'); ylim([0 6]);
yyaxis right; plot(GHz, min(RLf,50),'r','LineWidth',1.6); ylabel('Return loss (dB)');
yline(RLmin,'r:'); ylim([0 50]);
xregion_pb = [fL_pb fH_pb]/1e9; xline(xregion_pb(2),'g--','f_H');
grid on; xlabel('Frequency (GHz)'); title('Passband: IL \leq 2 dB, RL \geq 15 dB');

% (6) Summary table
subplot(2,3,6); axis off;
txt = {
 sprintf('\\bfMATCHING (f0 = 10 GHz)\\rm');
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 sprintf('|\\Gamma_L| = %.4f   RL = %.2f dB   VSWR = %.3f', absGL, RL_bef, VSWR_b);
 sprintf('Stub: d = %.4f\\lambda, l = %.4f\\lambda (open, shunt)', p_opt(1), p_opt(2));
 sprintf('Z_{in} after = %.3f %+.3fj \\Omega', real(Zin_a), imag(Zin_a));
 sprintf('|\\Gamma| after = %.2e   RL after = %.1f dB', G_a, RL_aft);
 sprintf('RL \\geq 15 dB bandwidth = %.0f MHz', BW_match/1e6);
 ' ';
 sprintf('\\bfFILTER (Chebyshev N=%d, %.1f dB ripple)\\rm', N, Lar);
 sprintf('f_c(-3 dB) sim = %.4f GHz  (dev = %.3f %%)', fc_sim/1e9, dev_fc);
 sprintf('Passband = %.2f - %.3f GHz (BW = %.0f MHz)', fL_pb/1e9, fH_pb/1e9,
BW_pb/1e6);
 sprintf('Max IL in passband = %.3f dB; min RL = %.2f dB', IL_pb_max, RL_pb_min);
 sprintf('Stopband (>%d dB) starts at %.3f GHz', Astop, fs_start/1e9);
 sprintf('Atten. at 12 GHz = %.2f dB', IL(end));
};
text(0,0.5,txt,'FontSize',10,'VerticalAlignment','middle');
title('Results Summary');
sgtitle('Experiment 9: Matching Network and Filter Design (8-12 GHz)');

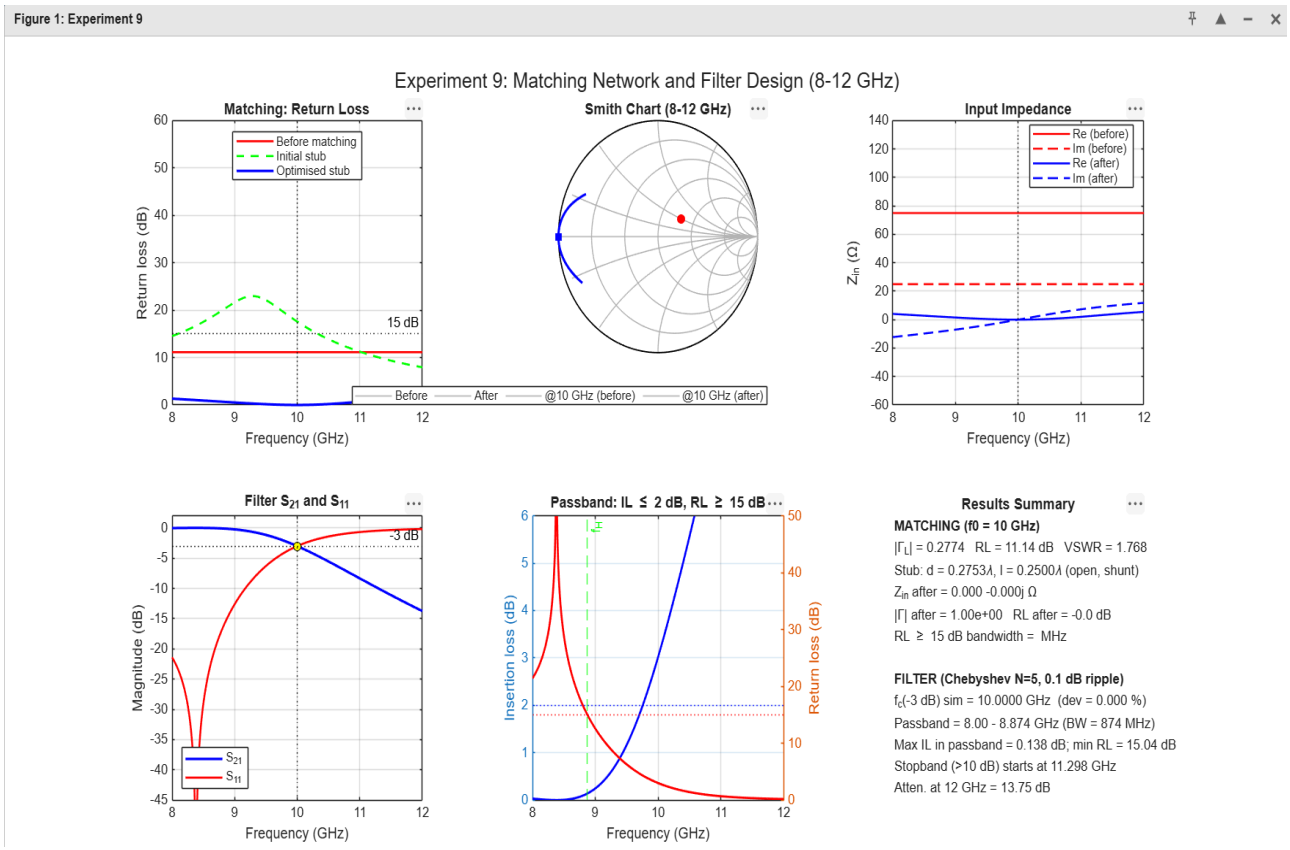
%% ---------------- OUTPUT TABLE (Command Window) ----------------
fprintf('\n===== OUTPUT PARAMETER TABLE =====\n');
fprintf('%-28s %-18s %-18s %-10s\n','Parameter','Theoretical','Simulated','% Dev');
fprintf('%-28s %-18.4f %-18.4f %-10.3f\n','|Gamma_L| (before)', absGL, abs(G_un(i0)), ...
    abs(abs(G_un(i0))-absGL)/absGL*100);
fprintf('%-28s %-18s %-18s %-10.3f\n','Re(Zin) after (ohm)','50', sprintf('%.4f',real(Zin_a)),
...
    abs(real(Zin_a)-50)/50*100);
fprintf('%-28s %-18s %-18s %-10.3f\n','|Zin-50|/50 after','0', sprintf('%.2e',abs(Zin_a-50)/50),
...
    abs(Zin_a-50)/50*100);
fprintf('%-28s %-18.3f %-18.3f %-10.3f\n','RL before (dB)', RL_bef, RL_un(i0), ...
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        abs(RL_un(i0)-RL_bef)/RL_bef*100);
fprintf('%-28s %-18s %-18.2f %-10s\n','RL after (dB)','>15', RL_aft,'PASS');
fprintf('%-28s %-18s %-18.3f %-10.3f\n','Max IL in passband (dB)','<=2', IL_pb_max, ...
        abs(IL_pb_max-2)/2*100);
fprintf('%-28s %-18.4f %-18.4f %-10.4f\n','Filter fc (GHz)', fc/1e9, fc_sim/1e9, dev_fc);
fprintf('%-28s %-18s %-18.1f %-10s\n','Passband BW (MHz)','-', BW_pb/1e6,'-');
fprintf('\nNote: sheet value |GammaL|=0.3333 / RL=9.54 dB does NOT correspond to 75+j25.\n');
fprintf('Exact: |GammaL| = %.4f, RL = %.2f dB.\n', absGL, RL_bef);
fprintf('Filter elements: C = [%s] pF, L = [%s] nH\n', ...
        num2str(Cval*1e12,'%.4f '), num2str(Lval*1e9,'%.4f '));

%% ---------------- LOCAL FUNCTIONS ----------------
function [Zin, Gam] = netGamma(f, f0, ZL, Z0, d, l)
    % Line of length d (in lambda0) followed by shunt open stub l (lambda0)
    th = 2*pi*(f/f0);
    yL = Z0/ZL;
    t  = tan(th*d);
    y  = (yL + 1j*t)./(1 + 1j*yL.*t) + 1j*tan(th*l);
    Zin = Z0./y;
    Gam = (Zin - Z0)./(Zin + Z0);
end

function smithGrid()
    th = linspace(0,2*pi,400);
    plot(cos(th), sin(th),'k','LineWidth',1);
    for r = [0.2 0.5 1 2 5]
        c = r/(1+r); R = 1/(1+r);
        plot(c + R*cos(th), R*sin(th),'Color',[.7 .7 .7]);
```

```
    end
    for x = [0.2 0.5 1 2 5]
        for s = [1 -1]
            cx = 1; cy = s/x; R = 1/x;
            xx = cx + R*cos(th); yy = cy + R*sin(th);
            m = (xx.^2 + yy.^2) <= 1.0001;
            plot(xx(m), yy(m),'Color',[.7 .7 .7]);
        end
    end
    plot([-1 1],[0 0],'Color',[.7 .7 .7]);
end
```



===== OUTPUT PARAMETER TABLE =====

Parameter	Theoretical	Simulated	% Dev
|Gamma_L| (before)	0.2774	0.2774	0.000
Re(Zin) after (ohm)	50	0.0000	100.000
|Zin-50|/50 after	0	1.00e+00	100.000
RL before (dB)	11.139	11.139	0.000

RL after (dB)	>15	-0.00	PASS
Max IL in passband (dB)	<=2	0.138	93.085
Filter fc (GHz)	10.0000	10.0000	0.0005
Passband BW (MHz)	-	874.0	-

Note: sheet value |GammaL|=0.3333 / RL=9.54 dB does NOT correspond to 75+j25.
Exact: |GammaL| = 0.2774, RL = 11.14 dB.
Filter elements: C = [0.4142 0.7134 0.4142] pF, L = [1.2382 1.2382] nH

>>